



## Comparison of linear beam theories

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### ABSTRACT

In this paper we consider three models for a cantilever beam based on three different linear theories: Euler–Bernoulli, Timoshenko and two-dimensional elasticity. Using the natural frequencies and modes as a yardstick, we conclude that the Timoshenko theory is close to the two-dimensional theory for modes of practical importance, but that the applicability of the Euler–Bernoulli theory is limited.

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## 1. Introduction

A beam is a three-dimensional object. Even the three-dimensional theory of elasticity is not perfect and further assumptions to simplify the theory must erode the extent to which it is applicable. In this paper we compare the Euler–Bernoulli theory to the Timoshenko theory and use the finite element method to compare the Timoshenko theory to a two-dimensional theory.

The Euler–Bernoulli theory for a beam originated in the 18th century. The effect of rotary inertia was introduced by Rayleigh in 1894. In 1921, Timoshenko proposed his theory where shear is also taken into account. Since shear and rotary inertia are ignored in the Euler–Bernoulli theory – see Section 2 – it is reasonable to assume that the Timoshenko theory is an improvement. Nevertheless, the assumption should be investigated.

To determine the validity of beam theories, comparisons can be made to the three-dimensional theory. One way of doing this is to consider phenomena due to wave motion. For the Euler–Bernoulli theory flexural waves of extremely short wave lengths are propagated at velocities approaching infinity, which is physically unreasonable (see [1, p. 321]).

The Euler–Bernoulli, Rayleigh and Timoshenko theories are compared to the three-dimensional theory in [2]. For this purpose a cylinder of infinite length free from lateral loading was considered. The partial differential equations which describe flexural (transverse) vibrations were derived by Pochhammer and Chree. The results are given in [3] and references are provided. Based on these equations, after early work was done by Bancroft [4], phase velocities for different wave lengths were calculated and published in [5]. In Appendix 3 of the paper [2], Davies calculated phase and group velocities and compared the results for the three “elementary theories” to those of the three-dimensional theory. The conclusion was that the Euler–Bernoulli and Rayleigh theories “...give results that are very wide of the truth...”, but “...the results derived from the Timoshenko theory are in excellent agreement with those deduced from the exact theory”.

As mentioned above, the conclusions above are based on the calculations in [5]. In [6] it is shown that these calculations are not correct, in the sense that it is assumed that flexural waves are propagated in only one mode. Based on the curves for phase velocity of flexural waves in [6], Fung [1, p. 324] concludes that “...the Timoshenko theory agrees reasonably well

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with the exact theory in the first mode, but wide discrepancy occurs in the second mode". In our view the impression is created that, although an improvement on the Euler–Bernoulli theory, the Timoshenko theory is far from satisfactory.

The Euler–Bernoulli theory for the transverse vibration of a beam is popular and still widely used. Although the Timoshenko theory is considered to be better, some authors, for instance Duva and Simmonds [7], do not agree that the Timoshenko theory is an unqualified improvement. According to [7], the corrections predicted by the Timoshenko theory are in some cases erroneous. They claim that for the first eigenfrequency of the cantilever beam, the Timoshenko theory provides a correction in the wrong direction and that this is due to "... effects at the built in end".

Considering the discussion above, a different approach, where the boundary conditions are taken into account, is called for. In this paper we consider different models for a cantilever beam and compare the natural frequencies and modes. To be specific, a two-dimensional model is used to evaluate the one-dimensional models. Judging by the results, the Timoshenko theory is superior. It was not unexpected that the verdict would be in favour of the Timoshenko theory, but the authors were surprised by the large numbers of accurate eigenvalues, even for relatively short beams.

In Section 2 we present the Euler–Bernoulli and Timoshenko models for a cantilever beam and compare the natural frequencies and modes in Section 3. A two-dimensional model for a cantilever beam is presented in Section 4. In Section 5 we derive the variational form for the eigenvalue problem and implement the finite element method in Section 6. In Section 7 the two-dimensional model is used to evaluate the one-dimensional models. In Section 8 we present a quantitative comparison between the eigenfunctions of the Timoshenko cantilever model and the two-dimensional model. At the end of Section 8 it is convenient to motivate the use of a two-dimensional theory to evaluate one-dimensional theories. Section 9 is the conclusion where we discuss comparison to the three-dimensional theory and suggest possible future research.

## 2. Beam theory

Consider a beam with constant density  $\rho$ , length  $\ell$  and cross sectional area  $A$ . The transverse displacement (deflection) of the cross section at  $x \in [0, \ell]$  at time  $t$  is denoted by  $w(x, t)$ . Assuming that plane cross sections remain plane, the rotation of a cross section is denoted by  $\phi(x, t)$ .

### Equations of motion

$$\rho A \partial_t^2 w = \partial_x V + P, \quad (1)$$

$$\rho I \partial_t^2 \phi = V + \partial_x M, \quad (2)$$

where  $I$  is the area moment of inertia,  $M$  the bending moment,  $V$  the shear force and  $P$  the transverse load (see [8, p. 331–337] and [9, p. 337]).

For the **Timoshenko theory** it is assumed that  $\partial_x w$  and  $\phi$  are small. The following constitutive equations for the moment  $M$  and the shear force  $V$  are used.

$$M = EI \partial_x \phi, \quad (3)$$

$$V = AG\kappa^2 (\partial_x w - \phi). \quad (4)$$

In these equations,  $E$  and  $G$  are elastic constants and  $\kappa^2$  the shear coefficient or shear correction factor. We refer the reader to [8, p. 337–338], [1, p. 323–324], [9, p. 337–338] and [10, p. 392–395].

Substituting the constitutive equations (3) and (4) into the equations of motion (1) and (2), yields the Timoshenko theory for the vibration of a beam.

$$\rho A \partial_t^2 w = \partial_x (AG\kappa^2 (\partial_x w - \phi)), \quad (5)$$

$$\rho I \partial_t^2 \phi = AG\kappa^2 (\partial_x w - \phi) + \partial_x (EI \partial_x \phi). \quad (6)$$

The partial differential equations above can be derived in different ways (see [1, p. 322–323] and [11]).

For the **Rayleigh theory** it is assumed that a cross section remains perpendicular to the neutral plane, i.e.  $\partial_x w = \phi$ . Eqs. (5) and (6) imply that

$$\rho A \partial_t^2 w - \rho I \partial_t^2 \partial_x^2 w = -\partial_x^2 (EI \partial_x^2 w).$$

The **Euler–Bernoulli theory** is obtained from the Rayleigh theory by simply ignoring the rotary inertia term.

### Dimensionless form

$$\text{Set } \tau = \frac{t}{t_0}, \quad \xi = \frac{x}{\ell}, \quad w^*(\xi, \tau) = \frac{w(x, t)}{\ell} \quad \text{and} \quad \phi^*(\xi, \tau) = \phi(x, t),$$

where  $t_0$  is to be specified. We introduce the dimensionless constants

$$\alpha = \frac{A\ell^2}{I}, \quad \beta = \frac{AG\kappa^2 \ell^2}{EI} \quad \text{and} \quad \gamma = \frac{\beta}{\alpha} = \frac{G\kappa^2}{E}.$$

The constant  $\gamma$  depends on the elastic constants and the shear correction factor  $\kappa^2$  that is determined by the shape of the cross section. The values of  $\kappa^2$  range between 1/2 and 1 (see [11] or [12, p. 173]). On the other hand, for isotropic materials it is assumed that  $\frac{G}{E} = \frac{1}{2(1+\nu)}$ , where  $\nu$  denotes Poisson's ratio (see [13, p. 174] or [1, Sec 7.2]). As a result, realistic values for  $\gamma$  range between 1/6 and 1/2. The constant  $\alpha$  is subject to significant variation. Note that  $\alpha = \frac{\ell^2}{r^2}$ , where  $r$  is the radius of gyration.

The forces and moments in dimensionless form are

$$P^*(\xi, \tau) = \frac{\ell P(x, t)}{G\kappa^2 A}, \quad V^*(\xi, \tau) = \frac{V(x, t)}{G\kappa^2 A} \quad \text{and} \quad M^*(\xi, \tau) = \frac{M(x, t)}{\ell G\kappa^2 A}.$$

A convenient choice for  $t_0$  is

$$t_0 = \ell \sqrt{\frac{\rho}{G\kappa^2}}.$$

Returning to the original notation we present the model problems for a **cantilever** beam.

### Timoshenko model

$$\begin{aligned} \partial_t^2 w &= \partial_x (\partial_x w - \phi), \\ \frac{1}{\alpha} \partial_t^2 \phi &= (\partial_x w - \phi) + \partial_x \left( \frac{1}{\beta} \partial_x \phi \right), \\ w(0, t) &= \phi(0, t) = 0, \\ \partial_x w(1, t) - \phi(1, t) &= \partial_x \phi(1, t) = 0. \end{aligned}$$

### Euler–Bernoulli model

$$\begin{aligned} \partial_t^2 w &= -\partial_x^2 \left( \frac{1}{\beta} \partial_x^2 w \right), \\ w(0, t) &= \partial_x w(0, t) = 0, \\ \partial_x^2 w(1, t) &= \partial_x^3 w(1, t) = 0. \end{aligned}$$

## 3. Comparison of one-dimensional models

To compare different models, we consider natural frequencies and modes. This is reasonable considering the fact that a solution of a vibration problem may be written as a series using eigenfunctions. In this section we compare the Euler–Bernoulli and Timoshenko theories. Firstly, it is convenient to use this case to motivate the approach. Secondly, the results can be used to determine limits for the applicability of the Euler–Bernoulli theory. In view of the results presented in Section 7, we have little doubt that the Timoshenko theory is closer to reality and may be used for this purpose.

### 3.1. Natural frequencies and modes

#### Euler–Bernoulli theory

To determine the natural frequencies and modes, the following eigenvalue problem must be solved.

#### Eigenvalue problem 1

$$\frac{1}{\beta} w^{(4)} = \lambda w \quad \text{with} \quad w(0) = w'(0) = w''(1) = w'''(1) = 0.$$

The eigenfunctions are

$$w_n(x) = A_n \sinh(\delta_n x) + B_n \cosh(\delta_n x) + C_n \sin(\delta_n x) + D_n \cos(\delta_n x)$$

where  $A_n, B_n, C_n$  and  $D_n$  are determined by the boundary conditions and  $\delta_n^4 = \beta \lambda_n$  (see e.g. [8, p. 333], [9, p. 335] or [14]). If  $\lambda_1, \lambda_2, \dots$  is the corresponding sequence of eigenvalues, then every solution of the vibration problem is given by

$$w(x, t) = \sum_{n=1}^{\infty} (P_n \cos \omega_n t + Q_n \sin \omega_n t) w_n(x), \quad (7)$$

where  $\omega_n = \sqrt{\lambda_n}$  are the natural angular frequencies. (The constants  $P_n$  and  $Q_n$  are determined by the initial conditions.)

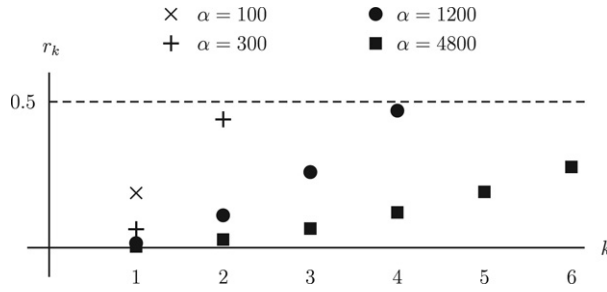


Fig. 1. Comparison of eigenvalues.

### Timoshenko theory

In this case we have the following eigenvalue problem.

#### Eigenvalue problem 2

$$\begin{aligned} -w'' + \phi' &= \lambda w, \\ -\frac{1}{\gamma} \phi'' - \alpha w' + \alpha \phi &= \lambda \phi, \end{aligned}$$

with the boundary conditions

$$\begin{aligned} w(0) &= \phi(0) = 0, \\ w'(1) - \phi(1) &= \phi'(1) = 0. \end{aligned}$$

Note that the eigenfunctions  $[w_n \ \phi_n]^T$  are vector valued. The component  $w_n$  that represents the deflection, is of the form

$$w_n(x) = A_n \sinh(\mu_n x) + B_n \cosh(\mu_n x) + C_n \sin(v_n x) + D_n \cos(v_n x),$$

where  $\mu_n$  and  $v_n$  are uniquely determined by  $\lambda_n$  (see [14]). For the deflection of the beam we have the representation (7), since the system of eigenfunctions is complete (see [15]).

### 3.2. Comparison

If the eigenvalues and eigenfunctions of the two models compare well, it is reasonable to expect that the solutions will compare well. Obviously the solutions will differ if the natural frequencies differ. As is well known, the Fourier coefficients decline rapidly and in many applications very few modes are needed. However, if the initial disturbance is highly localized, a substantial number of modes may be involved.

Denote the eigenvalues for the Euler–Bernoulli, Rayleigh and Timoshenko models by  $\lambda_{EB}^{(k)}$ ,  $\lambda_R^{(k)}$  and  $\lambda_T^{(k)}$  respectively. From numerical experiments we have  $\lambda_{EB}^{(k)} > \lambda_R^{(k)} > \lambda_T^{(k)}$  and  $\lambda_R^{(k)}$  is significantly closer to  $\lambda_{EB}^{(k)}$  than to  $\lambda_T^{(k)}$  (see e.g. [14]). We consequently do not consider the Rayleigh theory in the comparisons.

It is by now standard practice to compute eigenvalues and eigenfunctions using the finite element method, see e.g. [16, Chapter 6] (or [17] for the Timoshenko beam). However, we use the simple and accurate method in [14], whereby any chosen number of significant digits can be guaranteed. The results are presented in terms of the relative differences  $r_k$ , where  $r_k = (\lambda_{EB}^{(k)} - \lambda_T^{(k)}) / \lambda_T^{(k)}$ . These relative differences are presented graphically in Fig. 1. Note that for a beam with rectangular cross section, a value of  $\alpha = 1200$  represents a length to height ratio of 10:1, whereas  $\alpha = 300$  is associated with a length to height ratio of 5:1.

Due to the results in Section 7, we opine that the eigenvalues of the Timoshenko model involved in the present comparison, are reliable and the relative differences may be considered as relative errors. In Fig. 1 we see that the accuracy of the eigenvalues of the Euler–Bernoulli model depends on the parameter  $\alpha$  as well as the order of the mode. Note that  $r_4$  is almost 50% for  $\alpha = 1200$  against approximately 10% for  $\alpha = 4800$ .

When considering the application of the Euler–Bernoulli theory, one should decide on bounds for  $r_k$  that are acceptable. Using the formulae in [14], we calculated  $\delta_k$ ,  $\mu_k$  and  $v_k$  for  $\alpha \geq 1200$  and modes for which  $r_k < 0.05$ . We found that  $\delta_k \approx \mu_k \approx v_k$  and the relative errors for  $\delta_k$  compared to  $\mu_k$  and  $\delta_k$  compared to  $v_k$ , are less than  $r_k$ . For some initial conditions the Fourier coefficients decline rapidly and two modes are sufficient to approximate the solution for practical purposes. If this is the case, one may use the simpler Euler–Bernoulli theory.

### 4. Two-dimensional model for a cantilever beam

Consider a three-dimensional elastic body with density  $\rho$ . The displacement of a point  $\mathbf{x}$  in the reference configuration at time  $t$  is  $\mathbf{u}(\mathbf{x}, t)$ .

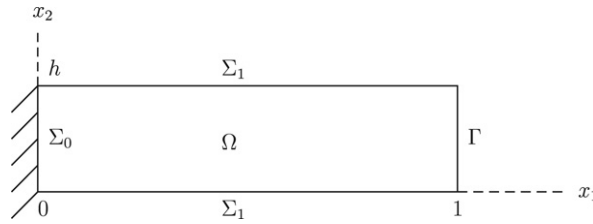


Fig. 2. Two-dimensional model.

### Equation of motion

$$\rho \partial_t^2 \mathbf{u} = \operatorname{div} T + \mathbf{Q},$$

where  $T$  is the first Piola stress tensor and  $\mathbf{Q}$  an external body force (density force) (see [1, Sec 5.5, 5.7] or [18, p. 125]).

In the case of small local displacements, the **infinitesimal theory of elasticity** or **linear elasticity** may be used. In this case the first Piola stress tensor is approximated by the Cauchy stress tensor (which is symmetric). For an explanation, see [18, p. 45–46, 122, 125]. Another explanation is given in [1, Sec 7.1]. In the matrix representation of  $T$ , the stress components are denoted by  $\sigma_{ij}$  and  $\operatorname{div} T$  is a vector with components

$$[\operatorname{div} T]_i = \partial_1 \sigma_{i1} + \partial_2 \sigma_{i2} + \partial_3 \sigma_{i3} \quad \text{for } i = 1, 2, 3.$$

Now consider a **prismatic** beam as illustrated in Fig. 2. We consider only “beam applications” where applied loads are symmetric with respect to the vertical plane through the center of the beam and the motion of the beam is parallel to the  $x_1 x_2$ -plane. Under these conditions it is reasonable to assume that the beam is in a state of plane stress. To be specific, we assume that  $\sigma_{3i} = \sigma_{i3} = 0$ . However, this does not imply that the problem is two-dimensional since  $\partial_3 \sigma_{ij}$  need not be zero. This is another assumption that we make. The interpretation is that the stresses are averages across the width of the beam. This approach is in line with Cowper’s [11] derivation of the Timoshenko theory. It is reasonable to assume that the two-dimensional theory is more accurate than one-dimensional theories (but obviously less accurate than a three-dimensional theory).

### Constitutive equations

The **infinitesimal strain**  $\varepsilon = (e_{ij})$  is given by

$$e_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i).$$

(See [18, p. 25] or [1, p. 155].)

We use **Hooke’s law for homogeneous isotropic materials** [1, Sec 9.1] or [13, p. 173, 182] for the special case of plane stress.

$$\begin{aligned} \sigma_{11} &= \frac{E}{1 - \nu^2} (e_{11} + \nu e_{22}), \\ \sigma_{22} &= \frac{E}{1 - \nu^2} (e_{22} + \nu e_{11}), \\ \sigma_{12} = \sigma_{21} &= \frac{E}{1 + \nu} e_{12}, \end{aligned}$$

where  $E$  is Young’s modulus and  $\nu$  Poisson’s ratio.

### Dimensionless form

The dimensionless variables and constants must be the same or compatible with those in Section 2. Set

$$\xi_i = \frac{x_i}{\ell}, \quad \mathbf{u}^*(\xi, \tau) = \frac{1}{\ell} \mathbf{u}(\mathbf{x}, t) \quad \text{and} \quad \sigma_{ij}^*(\xi, \tau) = \left( \frac{1}{G \kappa^2} \right) \sigma_{ij}(\mathbf{x}, t).$$

Returning to the original notation we present the equations of motion and constitutive equations in **dimensionless form**. To compute natural frequencies and modes, free vibration is considered, i.e.  $\mathbf{Q} = \mathbf{0}$ .

### Equation of motion

$$\partial_t^2 \mathbf{u} = \operatorname{div} T, \tag{8}$$

where

$$\operatorname{div} T = \begin{bmatrix} \partial_1 \sigma_{11} + \partial_2 \sigma_{12} \\ \partial_1 \sigma_{21} + \partial_2 \sigma_{22} \\ 0 \end{bmatrix}.$$

### Constitutive equations

$$\begin{aligned}\sigma_{11} &= \frac{1}{\gamma(1-\nu^2)} (\partial_1 u_1 + \nu \partial_2 u_2), \\ \sigma_{22} &= \frac{1}{\gamma(1-\nu^2)} (\partial_2 u_2 + \nu \partial_1 u_1), \\ \sigma_{12} &= \sigma_{21} = \frac{1}{2\gamma(1+\nu)} (\partial_1 u_2 + \partial_2 u_1).\end{aligned}\tag{9}$$

Now, consider a cantilever beam as in Fig. 2. The reference configuration  $\Omega$  is the rectangle given by  $0 \leq x_1 \leq 1$ ,  $0 \leq x_2 \leq h$  and the beam is fixed at  $\Sigma_0$  where  $x_1 = 0$ . (Note that  $h = \sqrt{12/\alpha}$ .) The parts of the boundary  $\Sigma_1$  (where  $x_2 = 0$  or  $h$ ) and  $\Gamma$  (where  $x_1 = 1$ ) are stress free. An obvious choice for the boundary condition on  $\Sigma_0$  is zero displacement (see e.g. [1, Section 7.7] or [13, Section 9.2]).

### Two-dimensional model problem

$$\begin{aligned}\partial_t^2 \mathbf{u} &= \operatorname{div} T \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \Sigma_0, \\ T \mathbf{e}_2 &= \mathbf{0} \quad \text{on } \Sigma_1, \\ T \mathbf{e}_1 &= \mathbf{0} \quad \text{on } \Gamma,\end{aligned}$$

with the constitutive equation given by (9).

### 5. Variational form for the two-dimensional model

Consider the equation of motion (8). Multiply both sides by an arbitrary vector valued function  $\phi$  and integrate over the reference configuration  $\Omega$ .

$$\iint_{\Omega} (\partial_t^2 \mathbf{u}) \cdot \phi \, dA = \iint_{\Omega} (\operatorname{div} T) \cdot \phi \, dA.$$

If  $T$  is symmetric,  $\operatorname{div} (T\phi) = (\operatorname{div} T) \cdot \phi + \operatorname{tr} (T\Phi)$ , where

$$\Phi = \begin{bmatrix} \partial_1 \phi_1 & \partial_2 \phi_1 \\ \partial_1 \phi_2 & \partial_2 \phi_2 \end{bmatrix}.$$

Application of the divergence theorem and the symmetry of  $T$  yield

$$\iint_{\Omega} \operatorname{div} (T\phi) \, dA = \int_{\partial\Omega} T\phi \cdot \mathbf{n} \, ds = \int_{\partial\Omega} T\mathbf{n} \cdot \phi \, ds.$$

Combining the results above, we have the **Green formula**

$$\iint_{\Omega} (\operatorname{div} T) \cdot \phi \, dA = - \iint_{\Omega} \operatorname{tr} (T\Phi) \, dA + \int_{\partial\Omega} T\mathbf{n} \cdot \phi \, ds.$$

Consequently,

$$\iint_{\Omega} (\partial_t^2 \mathbf{u}) \cdot \phi \, dA = - \iint_{\Omega} \operatorname{tr} (T\Phi) \, dA + \int_{\partial\Omega} T\mathbf{n} \cdot \phi \, ds\tag{10}$$

for any vector field  $\phi$  that is sufficiently smooth.

A bilinear form  $b(\mathbf{u}, \phi)$  is defined by

$$b(\mathbf{u}, \phi) = \iint_{\Omega} \operatorname{tr} (T\Phi) \, dA.$$

If Hooke's law, Eq. (9), is substituted into the bilinear form, we obtain

$$\begin{aligned}b(\mathbf{u}, \phi) &= \iint_{\Omega} (\sigma_{11} \partial_1 \phi_1 + \sigma_{12} \partial_1 \phi_2 + \sigma_{21} \partial_2 \phi_1 + \sigma_{22} \partial_2 \phi_2) \, dA \\ &= \frac{1}{\gamma(1-\nu^2)} \iint_{\Omega} (\partial_1 u_1 \partial_1 \phi_1 + \partial_2 u_2 \partial_2 \phi_2 + \nu(\partial_1 u_1 \partial_2 \phi_2 + \partial_2 u_2 \partial_1 \phi_1)) \, dA \\ &\quad + \frac{1}{2\gamma(1+\nu)} \iint_{\Omega} (\partial_1 u_2 + \partial_2 u_1) (\partial_1 \phi_2 + \partial_2 \phi_1) \, dA.\end{aligned}$$

The **space of test functions**  $T(\Omega)$  for the vibration problem consists of functions  $\phi$  with  $\phi_1$  and  $\phi_2$  in  $C^1(\bar{\Omega})$  and  $\phi = \mathbf{0}$  on  $\Sigma_0$ . Due to the boundary conditions,  $T\mathbf{n} \cdot \phi = \mathbf{0}$  on  $\Sigma_1$  and  $\Gamma$  and hence

$$\int_{\partial\Omega} T\mathbf{n} \cdot \phi \, ds = 0 \quad \text{for each } \phi \in T(\Omega).$$

### Variational form of the vibration problem

Find  $\mathbf{u}$  such that for  $t > 0$ ,  $\mathbf{u}(\cdot, t) \in T(\Omega)$  and

$$\iint_{\Omega} \partial_t^2 \mathbf{u} \cdot \phi \, dA = -b(\mathbf{u}, \phi) \quad \text{for each } \phi \in T(\Omega).$$

If  $\tilde{\mathbf{u}}(t, \mathbf{x}) = T(t)\mathbf{u}(\mathbf{x})$  is considered as a possible solution of the variational problem, it leads to the following eigenvalue problem.

### Eigenvalue problem 3

Find  $\mathbf{u} \in T(\Omega)$  and  $\lambda$  such that

$$b(\mathbf{u}, \phi) = \lambda \iint_{\Omega} \mathbf{u} \cdot \phi \quad \text{for each } \phi \in T(\Omega).$$

## 6. Galerkin approximation

To construct a finite-dimensional subspace  $S^h$  of  $T(\Omega)$ , consider a set of scalar valued functions  $\{\delta_1, \delta_2, \dots, \delta_p\}$  such that  $\delta_k = 0$  on  $\Sigma_0$  for each  $k$ . If a vector valued test function  $\phi$  is in  $S^h$ , each component is a linear combination of the chosen functions:

$$\phi_1(\mathbf{x}) = \sum_{j=1}^p \phi_{1j} \delta_j(\mathbf{x}) \quad \text{and} \quad \phi_2(\mathbf{x}) = \sum_{j=1}^p \phi_{2j} \delta_j(\mathbf{x}).$$

Note that  $S^h$  is spanned by the set with elements

$$[\delta_1 \, 0]^T, [\delta_2 \, 0]^T, \dots, [\delta_p \, 0]^T, [0 \, \delta_1]^T, [0 \, \delta_2]^T, \dots, [0 \, \delta_p]^T.$$

### Galerkin approximation

Find  $\mathbf{u}^h = [u_1^h \, u_2^h]^T \in S^h$  such that

$$b(\mathbf{u}^h, \phi) = \lambda \iint_{\Omega} \mathbf{u}^h \cdot \phi \, dA \quad \text{for each } \phi \in S^h. \quad (11)$$

In the rest of this section we use the notation  $(f, g) = \iint_{\Omega} fg$  to avoid clumsy expressions.

The following system of equations is obtained if  $\phi$  in (11) is first chosen as  $[\delta_i \, 0]^T$  for  $i = 1, 2, \dots, p$ , and then as  $[0 \, \delta_i]^T$  for  $i = 1, 2, \dots, p$ .

$$\begin{aligned} \frac{(\partial_1 u_1^h + \nu \partial_2 u_2^h, \partial_1 \delta_i)}{\gamma(1 - \nu^2)} + \frac{(\partial_1 u_2^h + \partial_2 u_1^h, \partial_2 \delta_i)}{2\gamma(1 + \nu)} &= \lambda(u_1, \delta_i) \\ \frac{(\partial_2 u_2^h + \nu \partial_1 u_1^h, \partial_2 \delta_i)}{\gamma(1 - \nu^2)} + \frac{(\partial_1 u_2^h + \partial_2 u_1^h, \partial_1 \delta_i)}{2\gamma(1 + \nu)} &= \lambda(u_2, \delta_i) \quad \text{for } i = 1, 2, \dots, p. \end{aligned} \quad (12)$$

This system is equivalent to (11).

For the approximate solution  $\mathbf{u}^h$  it holds that

$$u_1^h(\mathbf{x}) = \sum_{j=1}^p u_{1j} \delta_j(\mathbf{x}) \quad \text{and} \quad u_2^h(\mathbf{x}) = \sum_{j=1}^p u_{2j} \delta_j(\mathbf{x}).$$

The matrix formulation of the discrete eigenvalue problem is

$$\mathcal{K}\mathbf{u} = \lambda \mathcal{M}\mathbf{u},$$

with  $\mathbf{u} = [\mathbf{u}^1 \, \mathbf{u}^2]^T = [u_{11} \, u_{12} \, \dots \, u_{1p} \, u_{21} \, u_{22} \, \dots \, u_{2p}]^T$ . It is obtained by substituting the components of  $\mathbf{u}^h$  into the system of equations (12).

To construct  $\mathcal{M}$  and  $\mathcal{K}$ , we first define  $M$  and  $K^{rs}$  for  $r, s = 1, 2$  by

$$M_{ij} = (\delta_j, \delta_i) \quad \text{and} \quad K_{ij}^{rs} = (\partial_r \delta_j, \partial_s \delta_i) \quad \text{for } i, j = 1, 2, \dots, p.$$

**Table 1**  
Eigenvalues ( $\alpha = 1200$ )

| $j$ | $\lambda_{2D}^{(j)}$ | $k$ | $\lambda_{EB}^{(k)}$ | $\lambda_T^{(k)}$ | $\left  \frac{\lambda_{2D}^{(j)} - \lambda_{EB}^{(k)}}{\lambda_{2D}^{(j)}} \right $ | $\left  \frac{\lambda_{2D}^{(j)} - \lambda_T^{(k)}}{\lambda_{2D}^{(j)}} \right $ |
|-----|----------------------|-----|----------------------|-------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1   | 0.0317               | 1   | 0.0321               | 0.0316            | 0.0126                                                                              | 0.00315                                                                          |
| 2   | 1.14                 | 2   | 1.26                 | 1.14              | 0.105                                                                               | 0 <sup>a</sup>                                                                   |
| 3   | 7.72                 | –   | –                    | –                 | –                                                                                   | –                                                                                |
| 4   | 7.92                 | 3   | 9.90                 | 7.86              | 0.250                                                                               | 0.00758                                                                          |
| 5   | 26.2                 | 4   | 38.0                 | 25.9              | 0.450                                                                               | 0.0115                                                                           |
| 6   | 60.8                 | 5   | 104                  | 59.9              | 0.711                                                                               | 0.0148                                                                           |
| 7   | 69.3                 | –   | –                    | –                 | –                                                                                   | –                                                                                |
| 8   | 115                  | 6   | 232                  | 113               | 1.02                                                                                | 0.0174                                                                           |
| 9   | 192                  | 7   | 452                  | 188               | 1.35                                                                                | 0.0208                                                                           |
| 10  | 192                  | –   | –                    | –                 | –                                                                                   | –                                                                                |
| 11  | 291                  | 8   | 801                  | 284               | 1.75                                                                                | 0.0241                                                                           |

$\nu = 0.3$ ,  $\kappa^2 = 5/6$ ,  $\gamma = 0.3205$  (4 significant digits).

<sup>a</sup> Due to rounding.

Then

$$\mathcal{M} = \begin{bmatrix} M & 0 \\ 0 & M \end{bmatrix} \quad \text{and} \quad \mathcal{K} = \frac{1}{\gamma(1-\nu^2)} \begin{bmatrix} K^{11} + \left(\frac{1-\nu}{2}\right)K^{22} & \nu K^{21} + \left(\frac{1-\nu}{2}\right)K^{12} \\ \nu K^{12} + \left(\frac{1-\nu}{2}\right)K^{21} & K^{22} + \left(\frac{1-\nu}{2}\right)K^{11} \end{bmatrix}.$$

For  $\delta_i$  we choose the Hermite piecewise bicubic functions described in [16, p. 88], also called tensor-product Hermites, see [19, p. 66]. Recall that the test functions must satisfy the forced boundary conditions, and that only admissible test functions may be used. Care must be taken that the “not so obvious” test functions are also omitted. Since  $\phi_1 = \phi_2 = 0$  on  $\Sigma_0$ , the tangential derivatives  $\partial_2 \phi_1 = \partial_2 \phi_2 = 0$  on  $\Sigma_0$ .

In the process of refining the grid, we calculated the successive differences between two approximating sets of eigenvalues and used these differences to estimate the errors. (The MATLAB solver “eigs” was used on a grid of  $40 \times 10$  and the size of the eigenvalue problem was  $3564 \times 3564$ .) We are confident that the results are correct to three significant digits.

## 7. Comparison of eigenvalues

In this section we compare the eigenvalues for the two-dimensional beam to those of the Timoshenko and Euler–Bernoulli models. We considered  $\alpha = 1200$  ( $h = 0.1$ ) first, since this value is large enough to justify the application of a one-dimensional theory – from an intuitive point of view – while the eigenvalues of the Timoshenko and Euler–Bernoulli models differ substantially.

The results, correct to three significant digits, are given in Table 1. The eigenvalues of the two-dimensional model are denoted by  $\lambda_{2D}^{(k)}$ . Note that the value of  $\gamma$  is determined by  $\nu$ ,  $\kappa^2$  and  $\alpha$ .

Apart from the first two, the eigenvalues of the Euler–Bernoulli beam are obviously unacceptable. On the other hand, the eigenvalues for the Timoshenko beam compare well to those of the two-dimensional beam – all eight relative differences being less than 3%. It is notable that no eigenvalues of the Timoshenko model correspond to  $\lambda_{2D}^{(3)}$ ,  $\lambda_{2D}^{(7)}$  and  $\lambda_{2D}^{(10)}$ . We suspected that the “intruding” eigenvalues are not related to beam applications. A qualitative comparison of mode shapes confirmed this, see the examples below.

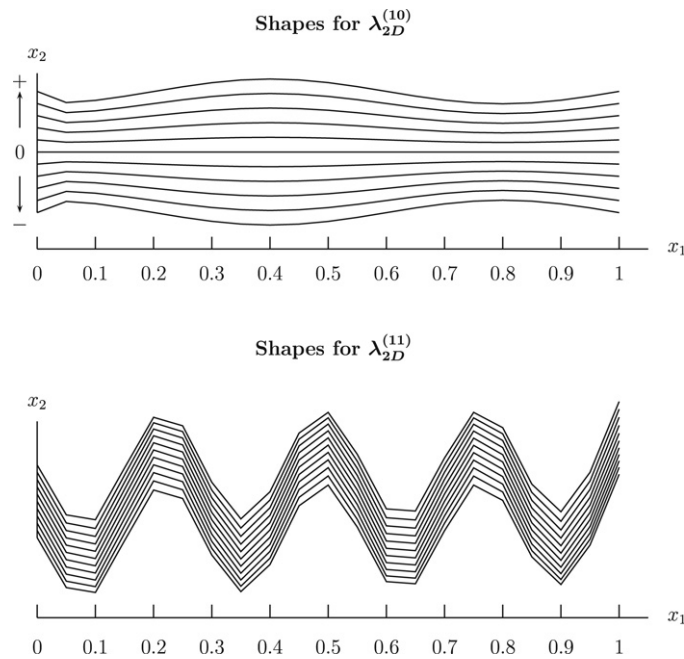
Suppose  $\mathbf{u}$  is an eigenfunction and we wish to investigate its shape. Referring (for convenience) to the  $x_1$ -direction as horizontal and the  $x_2$ -direction as vertical,  $u_1$  and  $u_2$  are the horizontal and vertical displacements of a point  $\mathbf{x}$  in the reference configuration. To determine whether an eigenfunction is related to a beam application, we consider the vertical displacements of lines that are horizontal in the reference configuration. The shapes of these lines are shown in Fig. 3 (not according to scale) for  $\lambda_{2D}^{(10)}$  and  $\lambda_{2D}^{(11)}$ . For each “graph” a discrete set of points 0.05 units apart, was used.

Note that for  $\lambda_{2D}^{(10)}$  the deflection of the neutral line (plane in reality) turns out to be zero and the lines at opposite sides of the neutral line are displaced in opposite directions. This clearly represents a two-dimensional effect not related to a beam application.

For  $\lambda_{2D}^{(11)}$  the situation is completely different. Here all the lines in the reference configuration have approximately the same displacement.

Having established the accuracy of the Timoshenko model for  $\alpha = 1200$ , it is natural to investigate the applicability of the model for “short” beams. The results for  $\alpha = 300$  are in Table 2.

The authors were surprised by the results in Table 2. In [14] it was shown that remarkable mode shapes correspond to eigenvalues (of the Timoshenko model) larger than  $\alpha$  and this led us to speculate that the relative differences for these eigenvalues will be significantly larger. The accuracy of eigenvalues number 9–14 is astonishing.

Fig. 3. Eigenfunctions ( $\alpha = 1200$ ).

**Table 2**  
Eigenvalues  $\alpha = 300$

| $j$ | $\lambda_{2D}^{(j)}$ | $k$ | $\lambda_T^{(k)}$ | $\lambda_{EB}^{(k)}$ | $j$ | $\lambda_{2D}^{(j)}$ | $k$ | $\lambda_T^{(k)}$ | $\lambda_{EB}^{(k)}$ |
|-----|----------------------|-----|-------------------|----------------------|-----|----------------------|-----|-------------------|----------------------|
| 1   | 0.122                | 1   | 0.121             | 0.129                | 13  | 397                  | 9   | 394               | 5290                 |
| 2   | 3.55                 | 2   | 3.51              | 5.05                 | 14  | 442                  | 10  | 440               | 8250                 |
| 3   | 7.73                 |     | –                 | –                    | 15  | 534                  |     | –                 | –                    |
| 4   | 20.2                 | 3   | 19.9              | 39.6                 | 16  | 539                  | 11  | 542               | <sup>a</sup>         |
| 5   | 56.1                 | 4   | 54.8              | 152                  | 17  | 596                  |     | –                 | –                    |
| 6   | 69.2                 |     | –                 | –                    | 18  | 603                  | 12  | 596               | <sup>a</sup>         |
| 7   | 114                  | 5   | 111               | 415                  | 19  | 658                  |     | –                 | –                    |
| 8   | 189                  |     | –                 | –                    | 20  | 717                  | 13  | 732               | <sup>a</sup>         |
| 9   | 193                  | 6   | 187               | 927                  | 21  | 748                  |     | –                 | –                    |
| 10  | 286                  | 7   | 278               | 1810                 | 22  | 783                  |     | –                 | –                    |
| 11  | 328                  | 8   | 330               | 3210                 | 23  | 787                  |     | –                 | –                    |
| 12  | 357                  |     | –                 | –                    | 24  | 790                  | 14  | 782               | <sup>a</sup>         |

$\nu = 0.3$ ,  $\kappa^2 = 5/6$ ,  $\gamma = 0.3205$  (4 significant digits).

<sup>a</sup> Extremely large values.

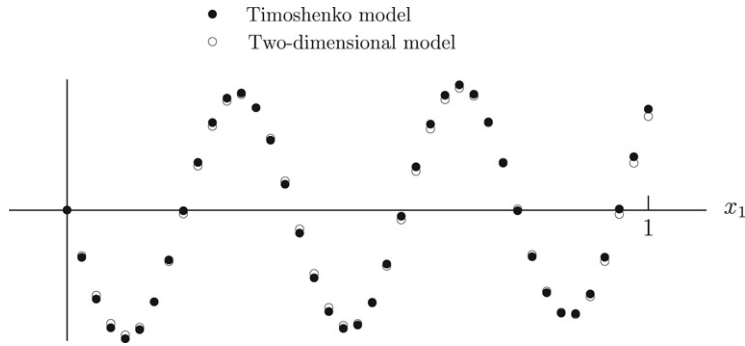
Even for  $\alpha = 100$  we found small relative differences for at least the first 10 comparable eigenvalues. In fact, if the eigenvalues are less than 100, the largest relative difference found was 3.22% for the 4th eigenvalue, which is the largest eigenvalue less than 100. For  $\lambda > 100$ , the largest relative difference for eigenvalues number 5–10 was found to be 4.63% (for eigenvalue number 9).

Considering the claim in [7] that  $\lambda_{EB}^{(1)}$  is more accurate than  $\lambda_T^{(1)}$ , note that the difference between  $\lambda_T^{(1)}$  and  $\lambda_{EB}^{(1)}$  is too small to be of practical importance unless  $\alpha$  is small. If, for example  $\alpha = 300$ , then  $\lambda_T^{(1)}$  is out by less than 1% while  $\lambda_{EB}^{(1)}$  is off the mark by approximately 5%. Except for the case where the eigenvalues of the three models are extremely close, it is hard to imagine that  $\lambda_{EB}^{(1)}$  could be closer to reality than  $\lambda_{2D}^{(1)}$ .

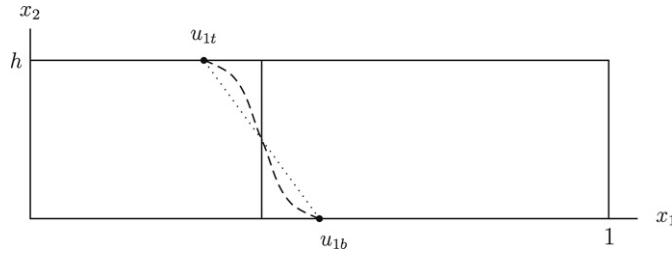
## 8. Quantitative comparison of eigenfunctions

How should the eigenfunctions of the Timoshenko model be compared to those of the two-dimensional model? Recall that the eigenfunction for the Timoshenko beam has two components. The component that represents deflection does not pose a problem as it may be compared to the transverse displacement of the neutral line.

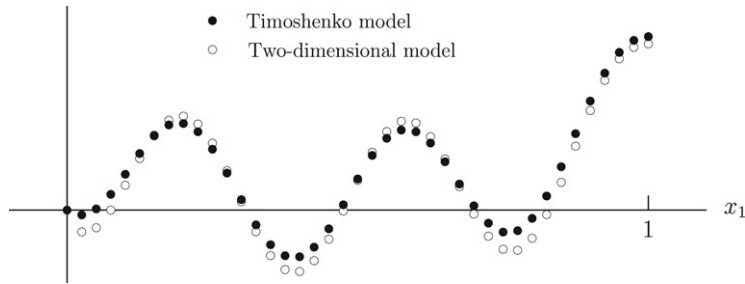
To compare the deflection for the Timoshenko theory against the deflection of the neutral line in the two-dimensional theory, a scale factor was calculated by considering the ratios of the deflections for both cases. (Recall that a multiple of an eigenfunction is again an eigenfunction.) As an example consider the comparable eigenvalues  $\lambda_{2D}^{(9)}$  and  $\lambda_T^{(6)}$  for  $\alpha = 300$ .



**Fig. 4.** Comparison of deflections ( $\alpha = 300$ ).  $\lambda_{2D}^{(9)} = 193$  and  $\lambda_T^{(6)} = 187$ .



**Fig. 5.** Average rotation of a cross section.



**Fig. 6.** Comparison of angles ( $\alpha = 300$ ).  $\lambda_{2D}^{(9)} = 193$  and  $\lambda_T^{(6)} = 187$ .

The mode shape of the Timoshenko model compares remarkably well with the neutral line in the two-dimensional model as shown in Fig. 4.

No scale is indicated on the vertical axis (in Fig. 4), since we consider eigenfunctions. In some cases the deflections of the Timoshenko theory and two-dimensional theory are so close that they appear to be the same.

It is not obvious what is to be compared to the angle  $\phi$  of the Timoshenko theory. In the Timoshenko theory it is assumed that a cross section remains plane while in reality it becomes warped (see Fig. 5). Now consider an eigenfunction for the two-dimensional model. Using the horizontal displacements of the (approximate) eigenfunction at the top ( $u_{1t}$ ) and at the bottom ( $u_{1b}$ ), we calculate an “average” gradient  $m_{2D} = \frac{u_{1t} - u_{1b}}{h}$ . Since  $u_{1t}$  and  $u_{1b}$  are components of an eigenfunction, the value of  $m_{2D}$  is not unique and may be large. This value of  $m_{2D}$  may be scaled to determine “small” values so that we may assume that  $m_{2D} = \tan \psi \approx \psi$ , providing us with an angle  $\psi$  suitable for comparison to  $\phi$ .

As for the displacements, scaling was necessary to make a comparison of the angles  $\phi$  and  $\psi$  possible. In Fig. 6 we see that the angles compare reasonably well.

From the discussion above, it is clear that comparison of a one-dimensional model to a two-dimensional model is problematic. It is evidently more so when a three-dimensional model is used. It is hard to imagine how either the Euler–Bernoulli model or the Timoshenko model can capture properties of a solution of a three-dimensional model and we opine that it is better to compare a two-dimensional model to a three-dimensional model as a separate investigation.

## 9. Conclusion

The results show that the Timoshenko model is remarkably accurate compared to the two-dimensional model, provided that the application is one for which beam theory is intended. However, comparison to a three-dimensional model is

desirable, but as mentioned before, a comparison of a two-dimensional model to a three-dimensional model is indicated. We envisage that a comparison of eigenvalues and eigenfunctions will be feasible.

As mentioned in Section 3, it is reasonable to expect that solutions of linear model problems will compare well if the eigenvalues and eigenfunctions compare well. However, what is really required is to determine the difference between the solutions that result from the same disturbance. This is clearly not a trivial matter and an investigation may be quite a challenge.

We have considered free vibration for a cantilever beam and compared the solutions for the Timoshenko and Euler–Bernoulli models, as indicated at the end of Section 3. If the parameter  $\alpha$  is relatively large and only a few modes are significant in a solution, then the difference between solutions for the two models is too small to be of practical importance. In our view a thorough comparison of the two models is worthy of further investigation.

The outcome of the investigation mentioned above will depend on the nature of the disturbance. This will determine the number of modes involved. If a large number of modes are necessary for an accurate approximation of the solution, it is likely that some values of the spatial derivative of the solution are large. If this is the case, the validity of any linear model is in doubt and comparisons should be made to nonlinear models.

A shortcoming of this paper is that damping is not considered. In some cases this does not matter, e.g. homogeneous viscous damping for the one-dimensional models. (The eigenvalue problem does not change.) In general the inclusion of damping will significantly complicate the investigation. Firstly, there are different types of damping of varying complexity (see e.g. [20] and [21]) and secondly, modal analysis may not be justified or even possible. Due to these complications damping was not considered.

## References

- [1] Y.C. Fung, *Foundations of Solid Mechanics*, Prentice-Hall Inc., Englewood Cliffs, New Jersey, 1965.
- [2] R.M. Davies, A critical study of the Hopkinson pressure bar, *Philosophical Transactions of the Royal Society, Series A* 240 (1948) 375–457.
- [3] A.E.H. Love, *A Treatise on the Mathematical Theory of Elasticity*, 4th edition, Cambridge University Press, 1927, Reprinted New York Dover (1963).
- [4] D. Bancroft, The velocity of longitudinal waves in cylindrical bars, *Physical Review* 59 (1941) 588–593.
- [5] G.E. Hudson, Dispersion of elastic waves in a solid circular cylinder, *Physical Review* 63 (1943) 46–51.
- [6] H.N. Abrahamson, Flexural waves in elastic beams of circular cross section, *Journal of the Acoustical Society of America* 29 (1957) 42–46.
- [7] J.M. Duva, J.G. Simmonds, The usefulness of elementary theory for the linear vibrations of layered, orthotropic elastic beams and corrections due to two-dimensional end effects, *Trans of ASME, Journal of Applied Mechanics* 58 (1991) 175–183.
- [8] H. Timoshenko, *Vibration Problems in Engineering*, second edition, D van Nostrand Company, Inc., New-York, 1937.
- [9] D.J. Inman, *Engineering Vibration*, Prentice-Hall Inc., Englewood Cliffs, New Jersey, 1994.
- [10] D.E. Newland, *Mechanical Vibration Analysis and Computation*, Longman, Essex, 1989.
- [11] G.R. Cowper, The shear coefficient in Timoshenko's beam theory, *Trans of ASME, Journal of Applied Mechanics* 33 (1966) 335–340.
- [12] A.P. Boresi, O.M. Sidebottom, F.B. Seely, J.O. Smith, *Advanced Mechanics of Materials*, Third edition, John Wiley & sons, New York, 1978.
- [13] N.O. Myklestad, *Statics of Deformable Bodies*, The Macmillan Company, New York, 1966.
- [14] N.F.J. van Rensburg, A.J. van der Merwe, Natural frequencies and modes of a Timoshenko beam, *Wave Motion* 44 (2006) 58–69.
- [15] M.A. Shubov, Asymptotic and spectral analysis of the spatially nonhomogeneous Timoshenko beam model, *Mathematische Nachrichten* 241 (2002) 125–162.
- [16] G. Strang, G.J. Fix, *An Analysis of the Finite Element Method*, Prentice-Hall, New Jersey, 1973.
- [17] L. Zietsman, N.F.J. van Rensburg, A.J. van der Merwe, A Timoshenko beam with tip body and boundary damping, *Wave Motion* 39 (3) (2004) 199–211.
- [18] R.J. Atkin, N. Fox, *An Introduction to the Theory of Elasticity*, Longman, London and New York, 1980.
- [19] G.F. Carey, J.T. Oden, *Finite Elements: A Second Course*, Volume II, Prentice-Hall, Inc., Englewood Cliffs, New Jersey, 1983.
- [20] D.L. Russell, in: G. Chen, J. Zhou (Eds.), *Vibration and Damping in Distributed Systems*, Volume II, CRC Presss, Boca Raton, Ann Arbor, London, Tokyo, 1993 (Chapter 5).
- [21] H.T. Banks, D.J. Inman, On damping mechanisms in beams, *ASME, Transactions, Journal of Applied Mechanics* 58 (1991) 716–723.